

THAW A CREATURE

THAW A CREATURE - CHAT/CODEX DEVELOPMENT HANDOFF - V0.9

Purpose: Continue development from the exact post-Day-5 checkpoint after the Jackpot Scale / Big-Number Progression design revision.

Project: Thaw A Creature - Roblox prototype / launch project

Primary workflow: ChatGPT designs/reviews one task -> precise Codex prompt -> Codex edits repo -> Studio runtime test through Rojo -> cleanup -> commit -> next task.

1. CANONICAL DOCUMENTS

Use these updated sources:

1. 01_Thaw_A_Creature_Concise_GDD_Canonical_v0.5.md
2. 02_Thaw_A_Creature_Technical_Implementation_3Week_v0.5.md
3. 03_Thaw_A_Creature_Bare_Minimum_Prototype_Technical_v0.9.md
4. 04_Thaw_A_Creature_Development_Plan_Canonical_v0.9.md
5. 05_Thaw_A_Creature_Implementation_Constants_Canonical_v0.9.md

Authority order: 1. GDD - player-facing intent. 2. Prototype Technical Spec - current prototype behavior. 3. Launch Technical Spec - production architecture. 4. Constants - exact first-pass numbers/content constraints. 5. Development Plan - task order/gates.

Do not silently reconcile conflicts. A newer version of the same document supersedes the older version.

2. PRODUCT NORTH STAR

Core fantasy:

Leave the Hearth, see mysterious Ice you desperately want, risk Warmth to retrieve it, bring it home, watch it thaw, CRACK -> CRACK -> SMASH, reveal a creature that changes progression, then go farther.

USP:

The collectible has a journey before it becomes a reward.

The v0.9 revision strengthens this with a pre-reveal jackpot signal:

Ice can spawn at absurd scale. The one visible size roll survives the entire journey and becomes the exact size lineage of the revealed creature copy.

Desired story:

“I saw this impossible giant Ice, barely got it home, and it became a massive creature with insane value.”

3. CURRENT CHECKPOINT - DAY 5 COMPLETE

Day 5 implementation is complete and runtime-verified:

- WorldCycleService / Great Frost countdown
- Frost presentation
- Great Frost Ice field reset
- active Great Frost 10x thaw boost
- Universal Survival Tool
- stationary Frost Sprite
- Guarded Ice encounters
- Equipped Creature Vitality / Frosted state
- minimal survival feedback
- Frost Sprite projectile collision fidelity fix

- Cash state + HUD
- 1-second aggregate Pen income
- Hearth upgrades

Survival Pressure Gate result:

PASS

Evidence: guarded Thick-Ice retrieval created a near-failure return with only 3 Warmth remaining; the player fought/rushed Sprites because of the Ice opportunity and cared most about the carried Ice + Warmth, not enemy farming.

Keep S01-S05 enabled. Do not expand enemy/combat scope automatically.

4. DAY-5 ECONOMY STATE BEFORE REVISION

Current implementation that exists in repo before Day-6 migration:

- Cash server-authoritative.
- Pen income currently ticks every 1 second.
- Existing rarity economy values were configured as:
 - Common 2/sec
 - Uncommon 5/sec
 - Rare 12/sec
 - Epic 20/sec
 - Legendary 28/sec
- Hearth currently has the initial small prototype levels from T08.

The v0.9 design **supersedes** the final source of Pen income and the Warmth/Hearth balance during Days 6-7:

- actual copy Pen income becomes species BaseIncome * size multiplier;
- actual Equipped Speed becomes species BaseSpeed * size multiplier;
- Hearth/Warmth are rebalanced into much larger numbers on Day 7.

Do not delete stable Day-5 behavior until its replacement task is implemented and verified.

5. NEW CANONICAL JACKPOT SCALE SYSTEM

Every wilderness Ice gets exactly one server-authoritative `WeightMultiplier`.

Formula:

```
ActualWeight = ReferenceWeight * WeightMultiplier
Scale = cbrt(ActualWeight / ReferenceWeight)
Scale = cbrt(WeightMultiplier)
```

Supported visual Scale:

```
0.75x minimum
5.00x maximum
```

First-pass size bands:

Small	8.0%	0.75-0.95x
Normal	72.0%	0.95-1.25x
Large	15.0%	1.25-1.75x
Huge	4.0%	1.75-2.50x
Colossal	0.9%	2.50-3.50x
Absurd	0.1%	3.50-5.00x

Within a band, bias toward the lower bound (`Random()`² interpolation in `WeightMultiplier` space) so near-5x results are much rarer than merely entering the Absurd band.

The exact `WeightMultiplier` is inherited unchanged:

```
wilderness Ice
-> carried Ice
-> deposited/thawing Ice
-> Ready Ice
-> reveal
-> creature copy
```

No second size roll at reveal. A giant Ice must produce a giant creature copy relative to that species' base size.

Extreme size is intentionally absurd/social. A 3.5-5x creature may cover much of the screen or visually spill across Pen spaces. Keep gameplay allocation clean and visuals non-blocking.

6. PER-COPY CREATURE ARCHITECTURE

The old species-count model is obsolete.

Target ownership:

```
CreatureCopies = {  
  [CreatureGUID] = {  
    CreatureID,  
    WeightMultiplier,  
  }  
}  
  
PenSlots[1..4] = CreatureGUID or nil  
EquippedCreatureGUID = CreatureGUID or nil  
DiscoveredCreatures[CreatureID] = true
```

Derived:

```
Scale = cbrt(WeightMultiplier)  
FinalIncome  
FinalSpeed
```

Inventory:

```
all CreatureGUIDs  
- Pen GUIDs  
- Equipped GUID
```

Species discovery remains species-based.

Only the inherited size roll is approved randomized copy variation. Do not add random Damage/Defense/levels/affixes.

7. SIZE-DERIVED STATS

Every species must eventually define:

```
ReferenceWeight  
BaseVisualScale  
BaseIncomePerSecond  
BaseWalkSpeed
```

Current first-pass formulas:

```
IncomeMultiplier = Scale ^ 2
FinalIncome = max(1, round(BaseIncomePerSecond * IncomeMultiplier))

SpeedMultiplier = 1 + 0.15 * (Scale - 1)
FinalWalkSpeed = BaseWalkSpeed * SpeedMultiplier
```

At 5x visual Scale: - Income $\sim$ 25x species base. - Speed $\sim$ 1.60x species base.

Income is intentionally much more explosive than Speed.

Creature presentation should expose:

```
SPEED <value>
PEN $<value>/s
```

An Equipped copy displays its intrinsic Pen stat but earns zero while Equipped.

8. CREATURE ASSET STATUS - D6-R00 COMPLETE

The live Studio audit verified exactly **11 direct-child Models** under `Workspace.Creatures` :

```
BEAR
HRAESVLGR
ICESERPENT
MAMMOTH
NIDHOGG
PENGUIN
RABBIT
SEAL
SLEIPNIR
TROLL
WOLF
```

Current configured roster:

```
Common
- Penguin
- Rabbit
- Seal

Uncommon
- Wolf
- Bear

Rare
```

```
- Mammoth  
  
Epic  
- Troll  
  
Legendary  
- Nidhogg
```

Currently unconfigured live models:

```
Sleipnir  
IceSerpent  
Hraesvlgr
```

Do **not** assign rarity, base stats, or Ice-pool placement to those three merely from this grouping. Final rarity/content placement still requires designer approval before D6-R05 .

Verified model-contract findings:

- all eleven lack `PrimaryPart` and a direct child named `Root` ;
- all eleven are simple two-MeshPart Models;
- there are no embedded gameplay scripts, Humanoids, AnimationControllers, Motor6Ds, or WeldConstraints;
- several have authored `GetScale() > 1.0` ;
- current runtime presentation normalizes collision/query/touch on clones.

Verified authored `GetScale()` values:

```
Penguin      1.0000  
Rabbit       1.0000  
Seal         1.0000  
Wolf         1.0000  
Bear         1.0000  
Mammoth      ~1.4373  
Troll        ~1.4798  
Sleipnir     ~1.4798  
Hraesvlgr    ~1.7404  
Nidhogg      ~2.1523  
IceSerpent   ~2.2605
```

These authored relative sizes are evidence for later species `BaseVisualScale` design. Future scale work must distinguish:

```
species base visual size  
x  
individual copy jackpot Scale
```

A `1.0x` copy means a normal-sized example of that species, not identical world dimensions across all species. Extreme `3.5x-5.0x` jackpot presentation remains intentional.

Production technical debt: The canonical creature asset source should eventually be reproducible through source control / build tooling. This does not block D6-R01.

9. BIG-NUMBER WARMTH DIRECTION

Warmth is now intentionally a satisfying large-number progression system.

First-pass revised constants:

```
Starting MaxWarmth = 100
Base Warmth Drain = 4/sec
Carry Multiplier = 2.00x
Hearth refill = 50% of MaxWarmth/sec
Low = 35% MaxWarmth
Critical = 15% MaxWarmth
```

Regions:

Snowfield	Regular	Recommended 100	Cold 1x
Frozen Pass	Thick	Recommended 1,000	Cold 10x
Ancient Expanse	Ancient	Recommended 10,000	Cold 100x
Black Ice Hollow	Black	Recommended 100,000	Cold 1,000x
Meteor Reach	Meteor	Recommended 1,000,000	Cold 10,000x

Warmth drain:

```
BaseDrain * RegionColdMultiplier * CarryMultiplier(if carrying)
```

Recommended Warmth is guidance only. Never gate entry.

10. REVISED HEARTH DIRECTION

The Hearth keeps exactly the same progression function but must become visually central.

First-pass Day-7 expansion:

Lv1	Start	+10 Warmth/sec
Lv2	\$150	+50/sec
Lv3	\$1,000	+250/sec
Lv4	\$7,500	+1,250/sec

Lv5	\$60,000	+6,250/sec
Lv6	\$500,000	+31,250/sec

Presentation target: - obvious fire/light/smoke/embers; - visible WarmZone; - HEARTH LV. X; - TRAINING +X WARMTH/s; - next cost / MAX.

No direct fixed MaxWarmth purchase. No cap.

11. REVISED DAY 6 TASK ORDER

Use a NEW Codex conversation for each numbered task unless fixing/cleaning that exact task.

1. D6-R00 - Audit the 11 Existing Creature Assets
2. D6-R01 - Per-Copy Creature Ownership Migration
3. D6-R02 - Authoritative Ice Weight + Scale Roll
4. D6-R03 - Ice Visual Scaling Through Full Journey
5. D6-R04 - Creature Size Inheritance
6. D6-R05 - Configure All 11 Existing Creatures - only after explicit base-stat/content approval
7. D6-R06 - Size-Based Income + Speed
8. D6-R07 - GUID Pen / Equip / Inventory
9. D6-R08 - Copy-Specific Pen Income
10. D6-R09 - Creature Stat Presentation
11. D6-R10 - Jackpot Validation Gate

Goal:

```
giant Ice
-> risky retrieval
-> same-scale giant creature
-> unusually strong Income
-> noticeably better Speed
```

12. REVISED DAY 7 TASK ORDER

Day 7 is an implementation day, not QA-only.

1. D7-R01 - Big-Number Warmth Rebalance
2. D7-R02 - Scalable Hearth Refill

3. D7-R03 - Hearth Progression Expansion
4. D7-R04 - Hearth Prominence Pass
5. D7-R05 - Expanded Progression Regions
6. D7-R06 - Recommended Warmth Signs
7. D7-R07 - Regional Cold Severity
8. D7-R08 - Ancient Ice
9. D7-R09 - Black Ice
10. D7-R10 - Meteor Ice if stable/time permits
11. D7-R11 - 11-Creature Pool Expansion
12. D7-R12 - Low-Warmth Feedback
13. D7-R13 - Progression Validation Gate

Continue sanity testing inside every implementation task.

13. WORKFLOW / CODEX SETTINGS

New implementation / architecture / nontrivial bug:

Model: GPT-5.6 Sol

Intelligence: Medium

Skill: Reference/Skills/roblox-systems-scripter.md

Codex conversation: NEW for each numbered task

Cleanup after verified runtime:

Model: GPT-5.6 Terra

Intelligence: Low

Skill: none

Codex conversation: SAME task conversation

Order:

1. ChatGPT gives one task prompt.
2. Codex implements; no commit.
3. Review diff/report.
4. Studio runtime test.
5. Fix/diagnose in same task conversation if needed.
6. Remove temporary diagnostics.
7. Sanity test.

8. Commit.
9. Confirm clean tree.
10. Next numbered task in NEW conversation.

Never trust Command Bar `require()` as proof of live state; test through actual gameplay/remotes/server lifecycle.

14. IMMEDIATE NEXT STEP

D6-R00 - Audit the 11 Existing Creature Assets is complete.

Start:

D6-R01 - Per-Copy Creature Ownership Migration

D6-R01 must migrate the current species-count allocation model to exact `CreatureGUID` copies before persistence work. Do not assign invented species baselines or rarity/pool placement for Sleipnir, IceSerpent, or Hraesvlgr during D6-R01.